

Euclidean Geometry In Mathematical Olympiads

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Problem 1.39 CGMO 2012 P5



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#8

Solution

By incenter-excenter O is on (ABC) but that doesn't really matter. Now observe $AD = AE$ by two tangent and $\angle DAO = \angle EAO$ since angle bisector. Then, we know that $\triangle DAO \cong \triangle EAO$ so $\angle AEO = \angle AEO$ and thus $\angle ODB = \angle OEC$.

Zanderman

4375 posts

Mar 16, 2021, 1:59 PM • 2

#10

I can't believe this problem.

I spent like 3 days completely stuck on it. -- I instantly noticed the Incenter-Excenter lemma, but then missed the triangle congruence. And then today after school, I spent like a second on it and saw it lol.

Bratin_Dasgupta

598 posts

Aug 14, 2021, 4:14 AM

#11



AlastorMoody wrote:

 $A - I - O,$

What does this mean??

ike.chen

1162 posts

Aug 16, 2021, 3:17 PM

#12



Bratin_Dasgupta wrote:



AlastorMoody wrote:

 $A - I - O,$

What does this mean??

That (non-standard) notation means A, I, O are collinear.

618173

1751 posts

May 15, 2022, 2:58 PM

#13

Clean!

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